What is the difference between a passive star and an active repeater in a fiber network?

A passive star has no electronics. The light from one fiber illuminates a number of others. An active repeater converts the optical signal to an electrical one for further processing.

Subnetting vs Supernetting

Subnetting is the process of dividing an IP network in to sub divisions called subnets whereas, Supernetting is the process of combining several IP networks with a common network prefix. Supernetting will reduce the number of entries in a routing table and also will simplify the routing process. In subnetting, host ID bits (for IP addresses from a single network ID) are borrowed to be used as a subnet ID, while in supernetting, bits from the network ID are borrowed to be used as the host ID.

What is Subnetting?

Process of dividing an IP network in to sub divisions is called subnetting. Subnetting divides an IP address in to two parts as the network (or routing prefix) and the rest field (which is used to identify a specific host). CIDR notation is used to write a routing prefix. This notation uses a slash (/) to separate the network starting address and the length of the network prefix (in bits). For example, in IPv4, 192.60.128.0/22 indicates that 22 bits are allocated for the network prefix and the remaining 10 bits are reserved for the host address. In addition, routing prefix can also be represented using the subnet mask. 255.255.252.0 (11111111.11111111.11111100.00000000) is the subnet mask for 192.60.128.0/22. Separating the network portion and the subnet portion of an IP address is done by performing a bitwise AND operation between the IP address and the subnet mask. This would result in identifying the network prefix and the host identifier.

What is Supernetting?

Supernetting is the process of combining several IP networks with a common network prefix. Supernetting was introduced as a solution to the problem of increasing size in routing tables. Supernetting also simplifies the routing process. For example, the subnetworks 192.60.2.0/24 and 192.60.3.0/24 can be combined in to the supernetwork denoted by 192.60.2.0/23. In the supernet, the first 23 bits are the network part of the address and the other 9 bits are used as the host identifier. So, one address will represent several small networks and this would reduce the number of entries that should be included in the routing table. Typically, supernetting is used for class C IP addresses (addresses beginning with 192 to 223 in decimal), and most of the routing protocols support supernetting. Examples of such protocols are Border Gateway Protocol (BGP) and Open Shortest Path First (OSPF). But, protocols such as Exterior Gateway Protocol (EGP) and the Routing Information Protocol (RIP) do not support supernetting.

IPv4 vs IPv6

(1) IPv4 is 32bit address space where as IPv6 has 128bit address space.
(2) CIDR was introduced for optimized usage of IPv4
(3) IPv4 format is four Octect and IPv6 is 8 block Hexadecimal.
(4) Even though IPv4 supports limited multicast, IPv6 is extensively supporting Multicast
(5) IPv6 avoid triangular routing, since it supports Mobility
(6) IPv6 supports bigger payload than IPv4
(7) IP tunneling is used for IPv4 and IPv6 interconnection at the moment.

IPv6 Example:
2607:f0d0:1002:0051:0000:0000:0202:0004

Give two example computer applications for which connection-oriented service is appropriate. Now give two examples for which connections less service is best.

Answer:
File transfer, remote login, and video on demand need connection-oriented service. On the other hand, credit card verification and other point-of-sale terminals, electronic funds transfer, and many forms of remote database access are inherently connectionless, with a query going one way and the reply coming back the other way.